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#include <stdio.h>

#define SIZE 10 // m = number of memory locations

int hashTable[SIZE];

// Initialize hash table
void init() {
    int i;
    for (i = 0; i < SIZE; i++)
        hashTable[i] = -1;
}

// Insert key using linear probing
void insert(int key) {
    int index = key % SIZE;
    int startIndex = index;

    while (hashTable[index] != -1) {
        index = (index + 1) % SIZE;

        if (index == startIndex) {
            printf("Hash table is full\n");
            return;
        }
    }

    hashTable[index] = key;
    printf("Key %d inserted at address %d\n", key, index);
}

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// Display hash table

void display() {
    int i;
    printf("\nHash Table:\n");
    for (i = 0; i < SIZE; i++) {
        if (hashTable[i] != -1)
            printf("Address %d : %d\n", i, hashTable[i]);
        else
            printf("Address %d : EMPTY\n", i);
    }
}

int main() {
    int n, key, i;

    init();

    printf("Enter number of employee records: ");
    scanf("%d", &n);

    for (i = 0; i < n; i++) {
        printf("Enter 4-digit employee key: ");
        scanf("%d", &key);
        insert(key);
    }

    display();

    return 0;
}

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Enter number of employee records: 3
Enter 4-digit employee key: 234
Key 234 inserted at address 4
Enter 4-digit employee key: 1244
Key 1244 inserted at address 5
Enter 4-digit employee key: 1254
Key 1254 inserted at address 6

Hash Table:
Address 0 : EMPTY
Address 1 : EMPTY
Address 2 : EMPTY
Address 3 : EMPTY
Address 4 : 234
Address 5 : 1244
Address 6 : 1254
Address 7 : EMPTY
Address 8 : EMPTY
Address 9 : EMPTY

Process returned 0 (0x0)   execution time : 7.165 s
Press any key to continue.
```